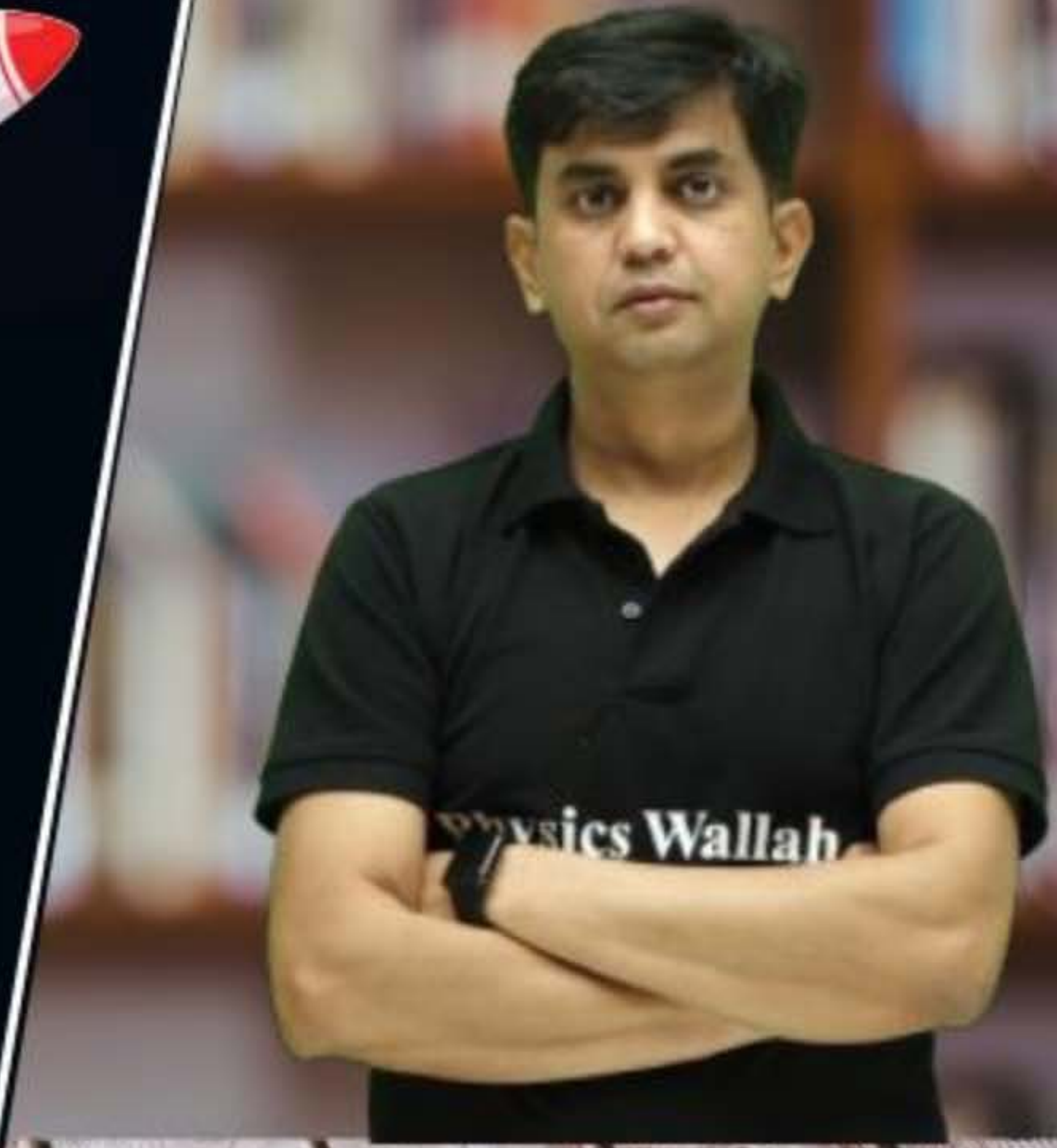
CS & IT ENGINEERING

Computer Network

IPv4 Address

Lecture No. - 13



By - Abhishek Sir

Recap of Previous Lecture

Topic ARP

Topic Supernetting

Topics to be Covered

Topic Supernetting

Topic NAT

ABOUT ME

Hello, I’m Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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[MSQ] [GATE-2024][1 Mark]

#Q. Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection.

Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

(A) The destination IP address is the IP address of R FALSE

(B) The destination IP address is the IP address of Y TRUE

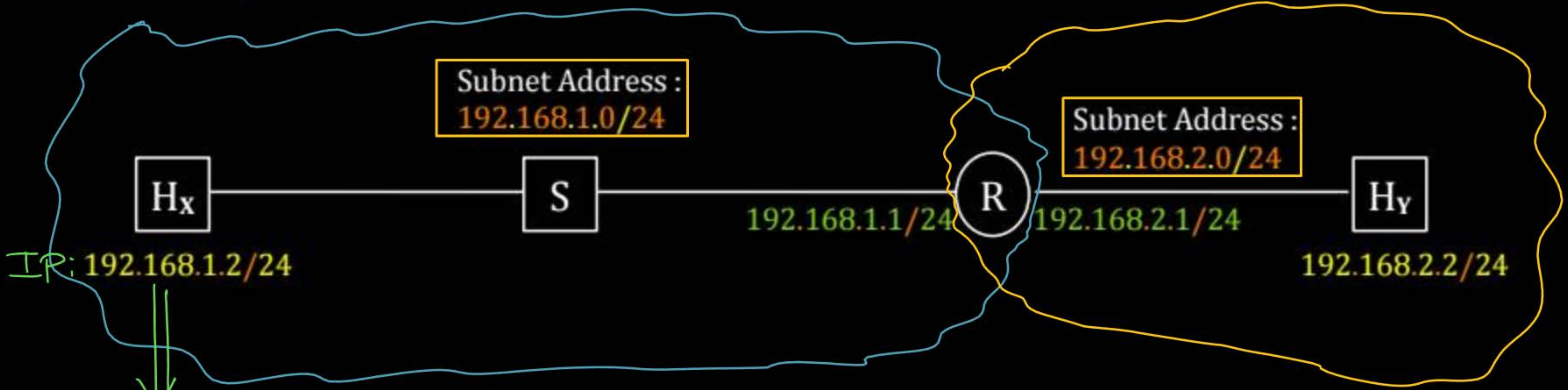
(C) The destination MAC address is the MAC address of S FALSE

(D) The destination MAC address is the MAC address of Y FALSE

Ans: B

Network Address : 192 . 168 . 0 . 0 / 16

With 8-bit subnetting



Default gateway

= 192.168.1.1

Source IP Address	:	192 . 168 . 1 . 2
Destination IP Address	:	192 . 168 . 2 . 2

Hx

H_Y

IP Packet

→ Host X (source host) finds destination host IP (Host Y) belongs to different subnet

→ Host X uses ARP Protocol to find MAC Address of default gateway [192 . 168 . 1 . 1]

→ Host X send frame to Router [The frame encapsulates the IP datagram]

Source MAC Address	:	Host X MAC Address
Destination MAC Address	:	Router MAC Address

$Frame_1$

[MCQ] [GATE-2019][2 Mark]

#Q. Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

(A) X sends an ARP request packet with broadcast IP address in its local subnet FALSE

(B) X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X

(C) X sends an ARP request packet with broadcast MAC address in its local subnet TRUE

(D) X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X

Ans: C

Topic : Supernetting

→ Prefix (Route) Aggregation / CIDR Aggregation

→ Combining (logically) smaller networks into single large network ⇒ Super-Net

→ Allow more efficient routing

→ Reduces the number of entries in routing table

→ All the Networks should be contiguous
[Block of addresses having contiguous prefixes]

Topic : Supernetting

Example 2 :- Consider following Network Addresses of networks. What should be supernet address ?

150 . 125 . 160 . 0 / 23

150 . 125 . 162 . 0 / 23

150 . 125 . 164 . 0 / 23

150 . 125 . 166 . 0 / 23

Supernet Address : 150·125·160·0/21

Topic : Supernetting

Example 2 :- 23 bit Prefix   9-bit HostID

$$\underbrace{150.125.10100\,\underline{00}}_{\text{23 bit Prefix}}\,\underbrace{0.00000000}_{\text{9-bit HostID}}/23$$

150 . 125 . 1 0 1 0 0 $\underline{0\,1}$ 0 . 0 0 0 0 0 0 0 0 / 23

150 . 125 . 1 0 1 0 0 $\underline{1\,0}$ 0 . 0 0 0 0 0 0 0 0 / 23

150 . 125 . 1 0 1 0 0 $\underline{1\,1}$ 0 . 0 0 0 0 0 0 0 0 / 23

000 . 00000000
⋮
001 . 11111111

010 . 00000000
⋮
011 . 11111111

100 . 00000000
⋮
101 . 11111111

110 . 00000000
⋮
111 . 11111111

Supernet Address : 150 . 125 . 160 . 0 /

$$\underbrace{150.125.10100}_{\text{Prefix(21 bit)}}\,\underbrace{00\,0.00000000}_{\text{11 bit HostID}}/21$$

Example 3 :- Consider following Network Addresses of networks.

210 . 192 . 0 . 0 / 13

210 . 200 . 0 . 0 / 13

210 . 208 . 0 . 0 / 13

210 . 216 . 0 . 0 / 13

Supernet Address : 210.192.0.0/11

Topic : Supernetting

Example 3 :-

$$\underbrace{\overbrace{210.110\;00}^{13\text{bit prefix}}}\;\overbrace{000.00000000.00000000}^{19\text{ bit Host ID}}/13$$

$$210.110\;01\;000.00000000.00000000/13$$

$$210.110\;10\;000.00000000.00000000/13$$

$$210.110\;11\;000.00000000.00000000/13$$

Supernet Address : 210.192.0.0 / 11

$$\underbrace{210.110}_{11\text{bit prefix}}\;\underbrace{00\;000.00000000.00000000}_{21\text{bit HostID}}/11$$

Topic : Supernetting

[H.W.]

Example 4 :- Consider following Network Addresses of networks.

10 . 90 . 0 . 0 / 16

10 . 90 . 64 . 0 / 18

10 . 90 . 192 . 0 / 18

Supernet Address :

Topic : IPv4 Address

→ Solution for IPv4 address (32-bits) range problem.

1. Network Address Translation (NAT)
   [Short-term solution]

2. IPv6 address (128 bits)
   [Permanent solution]

Topic : NAT

→ NAT : Network Address Translation

→ Internet : Public Network

→ Every network is considered as a private network

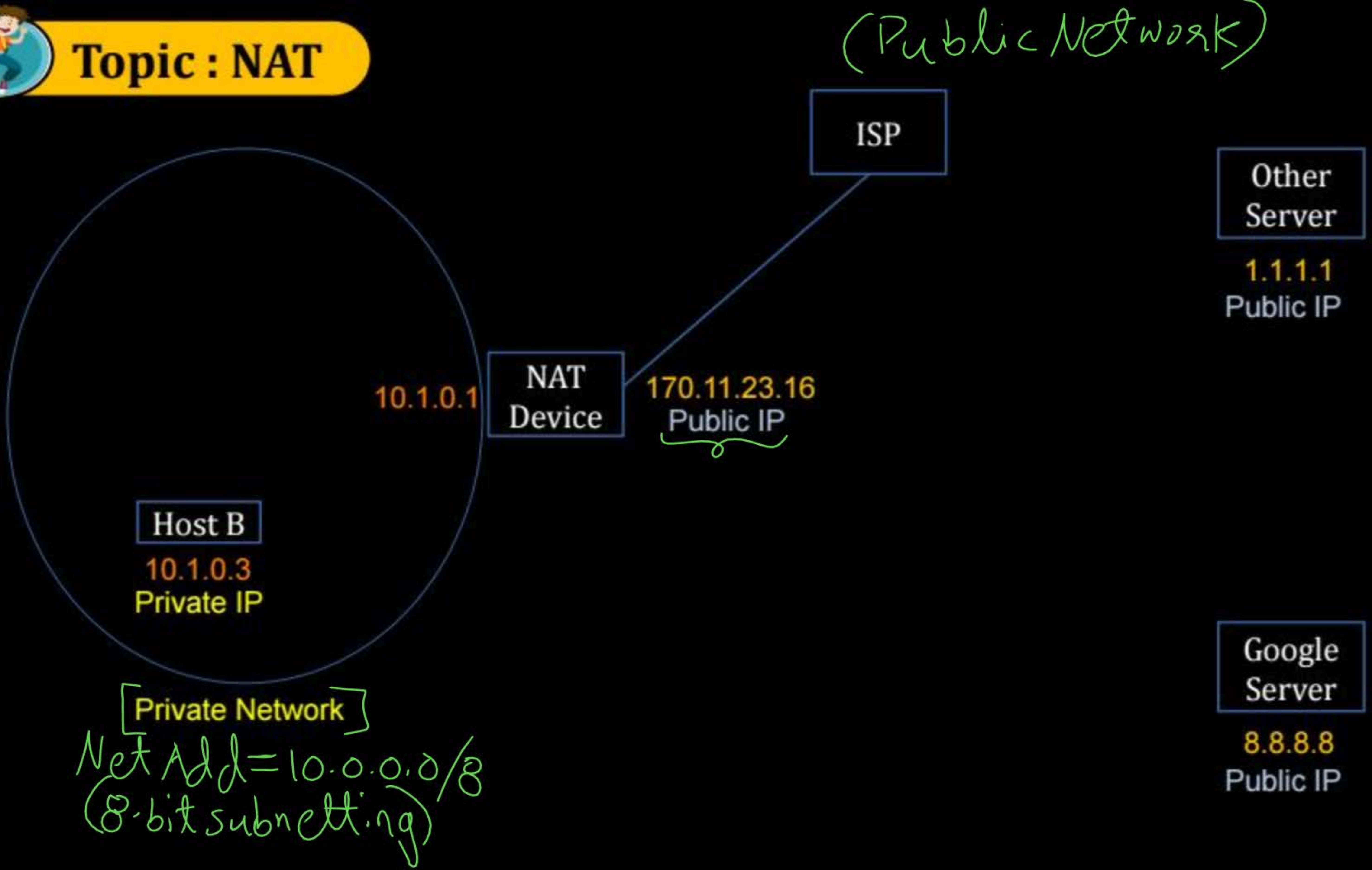
Topic : NAT

→ Every connected network is identified by unique public IPv4 Address [Assigned by ISP]

→ Total number of network can be exist (world wide) is $\boxed{2^{32}}$

→ All hosts inside a network is identified by private IPv4 Address

Topic : NAT



(Public Network)
ISP
Other Server
1.1.1.1
Public IP
10.1.0.1
NAT Device
170.11.23.16
Public IP
Host B
10.1.0.3
Private IP
Private Network
Net Add = 10.0.0.0/8
(8-bit subnetting)
Google Server
8.8.8.8
Public IP

Topic : NAT

CASE I :- Packet is goint out from internal to external network

→ Host B is sending one IP packet to Google server

IP Packet (Private IP) (Public IP)

Source IP Address	:	~~10.1.0.3~~ → 170.11.23.16
Destination IP Address	:	8.8.8.8 (public IP)

Topic : NAT Table

→ NAT device maintain,
NAT table for address translation of incoming datagram

Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]
10.1.0.3	8.8.8.8

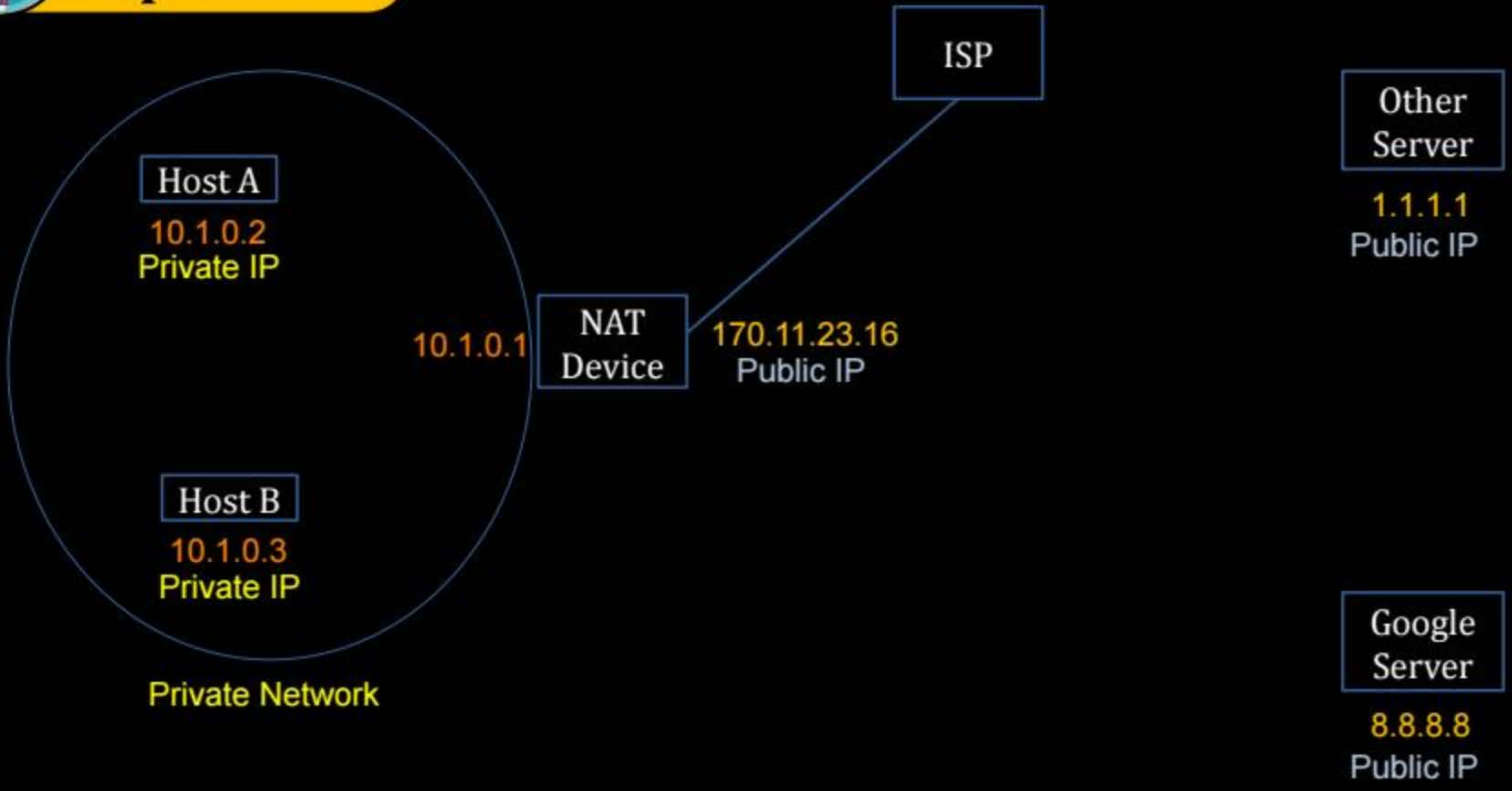
Topic : NAT

CASE II :- Packet is coming in from external to internal

→ Google server is sending one IP packet to Host B, in response of received packet

Source IP Address	:	8.8.8.8 (Public IP)
Destination IP Address	:	~~170.11.23.16~~ (Public IP) → 10.1.0.3 (Private IP)

Topic : NAT



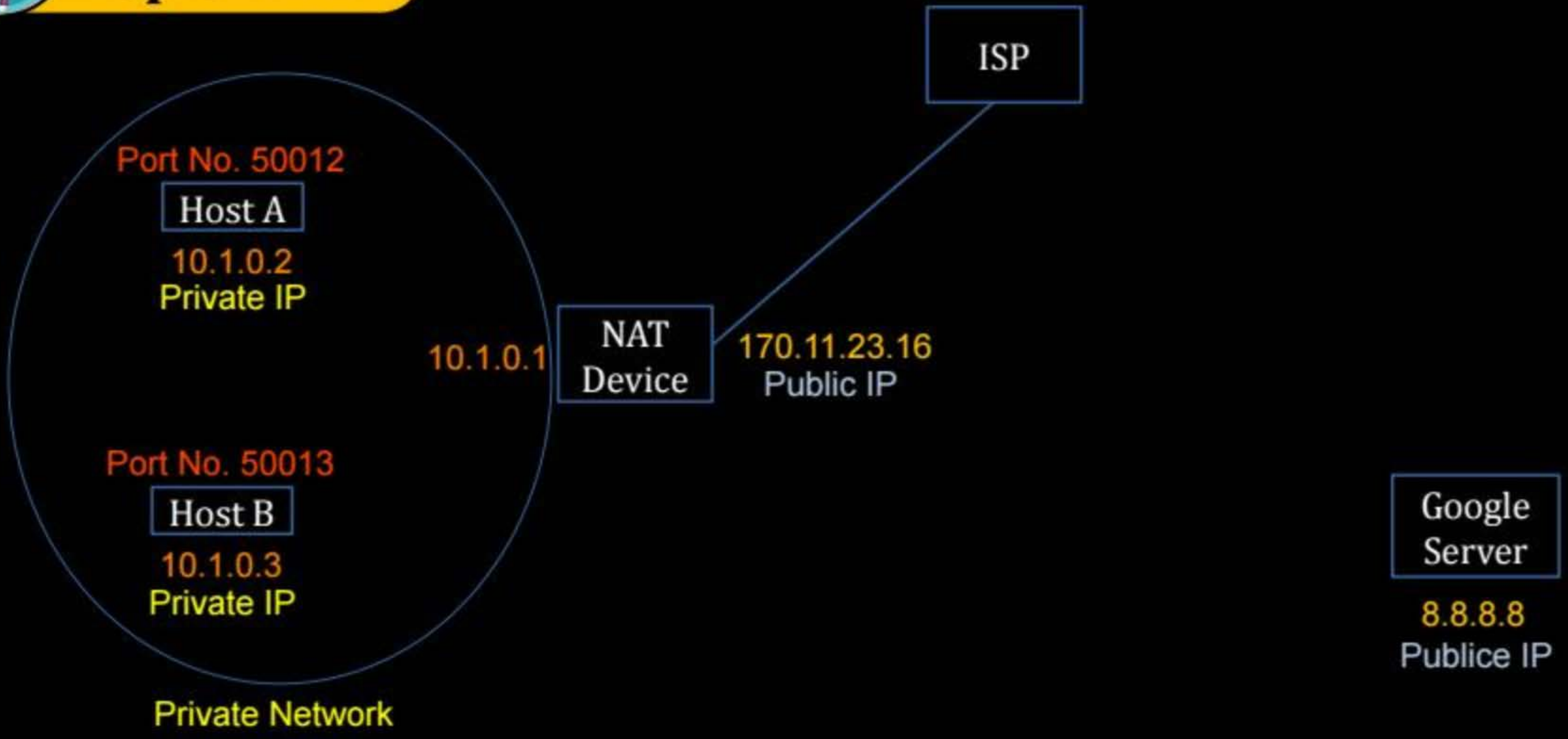
ISP
Other Server
1.1.1.1
Public IP
Host A
10.1.0.2
Private IP
10.1.0.1
NAT Device
170.11.23.16
Public IP
Host B
10.1.0.3
Private IP
Private Network
Google Server
8.8.8.8
Public IP

Topic : NAT Table

→ NAT device maintain,
NAT table for address translation of incoming datagram

Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]
10.1.0.2	1.1.1.1
10.1.0.3	8.8.8.8

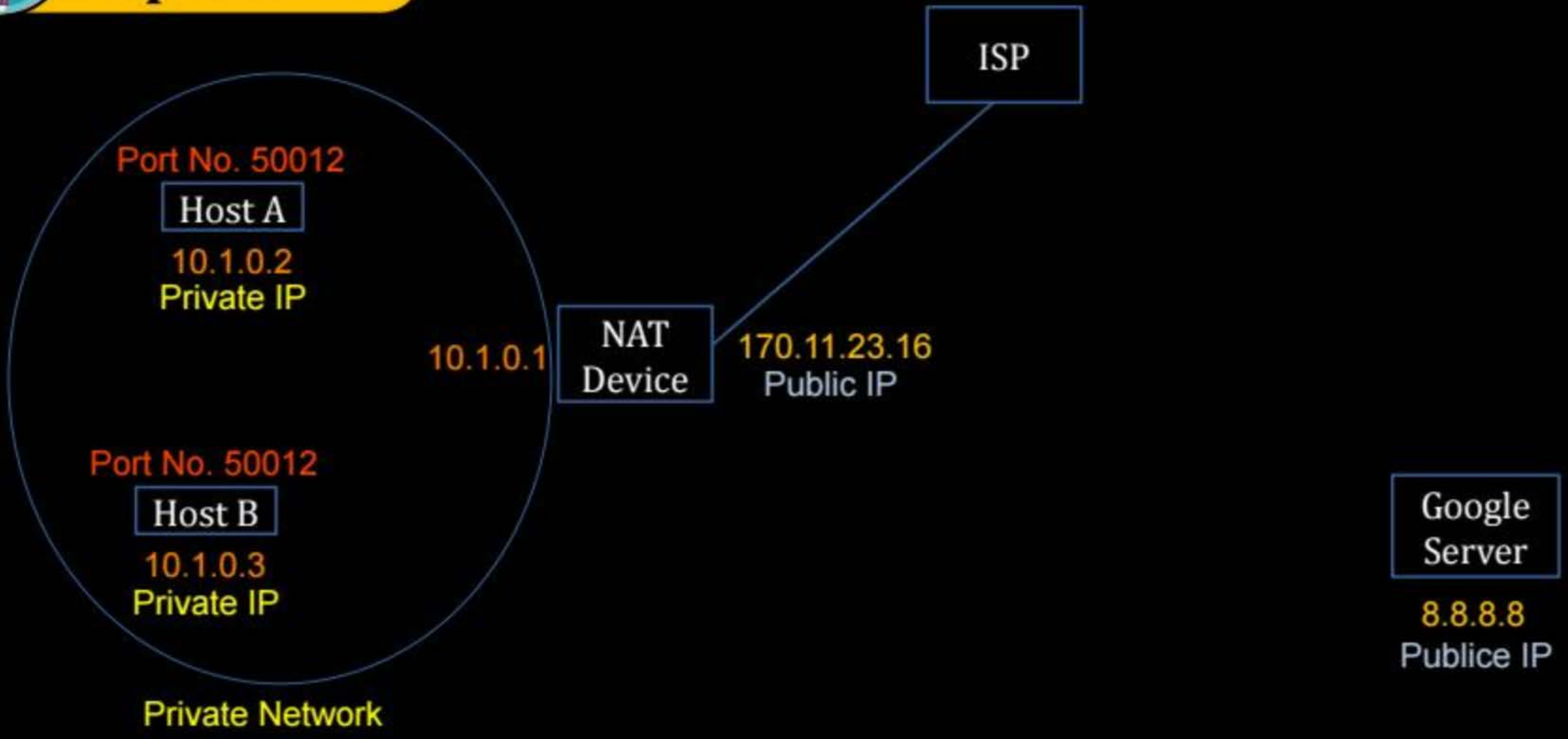
Topic : NAT


ISP
Port No. 50012
Host A
10.1.0.2
Private IP
10.1.0.1
NAT
Device
170.11.23.16
Public IP
Port No. 50013
Host B
10.1.0.3
Private IP
Google
Server
8.8.8.8
Publice IP
Private Network

Topic : NAT Table

Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]	Source Port Number
10.1.0.2	8.8.8.8	[50012]
10.1.0.3	8.8.8.8	[50013]

Topic : NAT


ISP
Port No. 50012
Host A
10.1.0.2
Private IP
10.1.0.1
NAT
Device
170.11.23.16
Public IP
Port No. 50012
Host B
10.1.0.3
Private IP
Google
Server
8.8.8.8
Publice IP
Private Network

Topic : NAT Table

Local Private IPv4 Address [Source IP Add.]	Global Public IPv4 Address [Destination IP]	Source Port Number	Modified Source Port Number
10.1.0.2	8.8.8.8	50012	50012
10.1.0.3	8.8.8.8	50012	50020

Topic : NAT Device

→ NAT device update address field of every outgoing and incoming IP packet

→ Mapping internal IP addresses and ports to external IP addresses and ports

Topic : NAT Device

→ For every outgoing datagram,
it modify Source IP address from private IP address to public IP address

V. Imp.

→ For every incoming datagram,
it modify Destination IP address from public IP address to private IP address

V. Imp.

→ As per requirement, NAT device can modify
Source Port Number field for outgoing packet
and Destination Port Number field for incoming packet

Topic : Private IPv4 Address

→ Network addresses for private IPv4 Networks :

$\boxed{10.0.0.0/8}$ → for college/org. (for large N/w)

172.16.0.0/12

$\boxed{192.168.0.0/16}$ → for Home (small network)

2 mins Summary

Topic Supernetting → (CS-2022) (one que)

Topic NAT → (CS-2024) (one que)

[DHCP → No any ques.]

THANK - YOU

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